CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of B. Tech. Examination (CE/IT/EC)
May 2014

CL 104 Basics of Civil Engineering

Date: 21.05.2014, Wednesday   Time: 01:30 p.m. To 04:30 p.m.   Maximum Marks: 70

Instructions:
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Explain the role of Civil Engineer. [04]

(b) Write the fundamental principles of Surveying and explain any one. [03]

Q - 2 (a) Enlist various instruments used in chaining and describe with sketch any two of them. [07]

(b) Explain the procedure of Reciprocal ranging. [05]

(c) Define surveying and leveling. [02]

OR

Q - 2 (a) Define offsetting. List down the instruments used for taking offsets and write the principle of Optical Square. [04]

(b) A 20 m chain was found to be 8 cm too long at the end of a day's work after measuring 4000 m. If the chain was correct before commencement of work, find the correct length of the line. [05]

(c) What is ranging? Explain direct ranging with neat sketch. [05]

Q - 3 (a) Given below are the bearings of lines of traverse ABCD. Find the included angles and apply necessary checks. [05]

Line	Fore Bearing
AB	N 45^0 E
BC	N 75^0 E
CD	S 35^0 W
DA	N 65^0 W

(b) Differentiate between Prismatic compass and Surveyor's compass. [03]

(c) The following staff readings were recorded in a leveling operation. [06]
1.175, 2.605, 1.835, 2.405, 1.145, 0.965, 1.115, 1.785, 1.225, 1.645 & 1.715. Station A is the benchmark with reduced level 175.68m. Find the RLs of all the other points by rise and fall method. The first reading was taken on point A and instrument was shifted after the 2^{nd}, 6th and 9^{th} readings. Apply the usual checks.

OR

Q - 3 (a) Explain with sketch temporary adjustment of a dumpy level. [05]

(b) Convert the following [05]

1) WCB to RB

(i) 285° (ii) 198°30' (iii) 58°

2) QB to WCB

(i) N23°W (ii) S82°E

(c) Define following. [04]

1. True Bearing 2. Magnetic Bearing 3. Reduced level 4. Bench mark.

SECTION – II

Q - 4 (a) i. Enlist different uses of cement. [02]

ii. Enlist properties of fresh Concrete. Write down the compressive strength of 53 grade cement at the end of 28 days. [02]

iii. Describe the characteristics of good bricks. [03]

Q - 5 (a) i. Write down difference between Load Bearing and Framed Structure. [05]

ii. Enlist Classification of Building based on Occupancy. [02]

(b) Draw a detailed sketch of a cross section of a wall showing the components of the building. [07]

OR

Q - 5 (a) Draw the chart showing classification of foundation and draw a sketch typical wall footing for 20 cm thick wall. [07]

(b) Explain the loads taken into consideration while designing a structure. [07]

Q - 6 (a) Enumerate various principles of planning and explain roominess and privacy in detail. [07]

(b) Write down functions of the following [07]

(a) Plinth (b) Sill (c) Beam (d) Roof (e) Weather shed (f) Door (g) Lintel

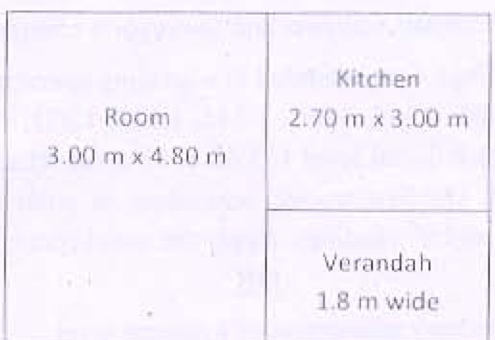
OR

Q - 6 (a) i. Write a short note on Setback and Built up area [05]

ii. Draw symbols for the following [02]

(a) Brick (b) Earth (c) Road bridge (d) Embankment

(b) Draw a detailed plan of a building from the line diagram. Assume suitable scale. Consider plinth height as 60 cm. Also prepare a schedule giving relevant description of doors and windows. All walls are 30 cm thick. [07]



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